Chapter 6. Conditional Heteroscedasticity Models

ARCH, GARCH, change points, Value-at-Risk

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6.1 Introduction

Motivation. Financial returns are nearly uncorrelated, yet their size is predictable, so the conditional variance (*volatility*) becomes the object of inference.

Volatility clustering means large moves come in bursts, i.e. the conditional variance depends on the past. For AR(1) $y_t = \phi y_{t-1} + \epsilon_t$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$,

$$E(y_t \mid y_{t-1}) = \phi y_{t-1}, \qquad \mathrm{Var}(y_t \mid y_{t-1}) = \sigma^2,$$

constant. With $y_t = 0.1 y_{t-1} + \epsilon_t$, $\epsilon_t \mid \epsilon_{t-1} \sim N(0,\, 1 + 0.5\epsilon_{t-1}^2)$, a large shock raises the next variance (Figure 1).

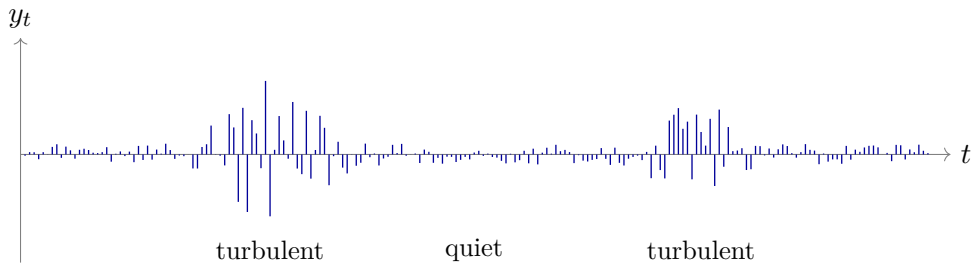


Figure 1: Volatility clustering: large moves come in bursts.

6.2 ARCH models

Motivation. ARCH makes the conditional variance a function of past squares, giving uncorrelated heavy-tailed data whose squares are AR.

With $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ (the book's I_{t-1}), we model

$$y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \qquad \sigma_t^2 = \sigma_t^2(y_{t-1}, y_{t-2}, \dots), \tag{6.1}$$

often with finitely many lags $y_{t-1}, \dots, y_{t-q}$.

6.2.1 ARCH(1)

Let $\{\xi_t\} \sim IID\, N(0,1)$ and

$$y_t = \sigma_t \xi_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2, \qquad \alpha_0 > 0,\ 0 \le \alpha_1 < 1, \tag{6.2}$$

with ξ_t independent of σ_t. Then

$$P(y_t \le x \mid y_{t-1} = y) = P(\xi_t \le x/\sqrt{\alpha_0 + \alpha_1 y^2}) = \Phi(x/\sqrt{\alpha_0 + \alpha_1 y^2}), \tag{6.3}$$

so $y_t \mid y_{t-1}, y_{t-2}, \cdots \sim N(0, \sigma_t^2)$: the *ARCH(1) model*, an instance of (6.1). Often only $\xi_t \sim IID(0,1)$ is assumed.

Iterating $y_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2$,

$$y_t^2 = (\alpha_0 + \alpha_1(\alpha_0 + \alpha_1 y_{t-2}^2)\xi_{t-1}^2)\xi_t^2 = \cdots = \alpha_0 \sum_{j=0}^{n} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2 + \alpha_1^{n+1} \xi_t^2 \cdots \xi_{t-n}^2 y_{t-n-1}^2, \tag{6.4}$$

which suggests

$$y_t^2 = \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2, \qquad \sigma_t^2 = \alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2, \tag{6.5}$$

i.e. the candidate solution

$$y_t = \xi_t \Big(\alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2 \Big)^{1/2}, \tag{6.6}$$

a nonlinear function of $\xi_t, \xi_{t-1}, \ldots$.

Proposition (Properties of ARCH(1), [Supplement])

Let $\alpha_0 > 0$, $0 \le \alpha_1 < 1$ and $\{\xi_t\} \sim IID(0,1)$.
(1) The series in (6.6) converges a.s.; $\{y_t\}$ is a strictly stationary solution of (6.2), σ_t^2 is a function of $\xi_{t-1}, \xi_{t-2}, \ldots$, and (σ_t^2, y_t) is independent of $\xi_{t+1}, \xi_{t+2}, \ldots$.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \mathrm{Var}(y_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and

$$Ey_t^2 = E\sigma_t^2 = \frac{\alpha_0}{1-\alpha_1}; \tag{6.7}$$

$\{y_t\}$ is a stationary white noise.
(3) If $\xi_t \sim N(0,1)$, then $Ey_t^4 < \infty$ if and only if $3\alpha_1^2 < 1$, and then the kurtosis is

$$\frac{Ey_t^4}{(Ey_t^2)^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3 \quad (\alpha_1 > 0). \tag{6.8}$$

(4) If $Ey_t^4 < \infty$, then $\nu_t = y_t^2 - \sigma_t^2$ is a white noise and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t, \tag{6.9}$$

so $\{y_t^2\}$ is an AR(1) process.

Proof.

Step 1. Let $h_t = \alpha_0 + \alpha_0\alpha_1 \sum_{j\ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2 \ge 0$. As $E(\xi_{t-1}^2 \cdots \xi_{t-j-1}^2) = 1$, $Eh_t = \alpha_0/(1-\alpha_1) < \infty$, so the series converges a.s., and

$$\alpha_0 + \alpha_1 \xi_{t-1}^2 h_{t-1} = \alpha_0 + \alpha_0\alpha_1 \xi_{t-1}^2 + \alpha_0 \sum_{j\ge 0} \alpha_1^{j+2} \xi_{t-1}^2 \xi_{t-2}^2 \cdots \xi_{t-j-2}^2 = h_t,$$

so $\sigma_t^2 = h_t$, $y_t = \sigma_t \xi_t$ satisfy $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$. Being a fixed function of the shifted i.i.d. sequence, $\{y_t\}$ is strictly stationary; independence is immediate.

Step 2. $\sigma_t = (\alpha_0 + \alpha_1 y_{t-1}^2)^{1/2}$ is $\mathcal{F}_{t-1}$-measurable and $\xi_t \perp \mathcal{F}_{t-1} \subset \sigma(\xi_{t-1}, \ldots)$, so $E(y_t \mid \mathcal{F}_{t-1}) = \sigma_t E\xi_t = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and $Ey_t^2 = Eh_t = \alpha_0/(1-\alpha_1)$. For $h \ge 1$, $E(y_t y_{t+h}) = E(y_t\, E(y_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$.

Step 3. $y_t^2 = \alpha_0 \sum_{j\ge 0} \alpha_1^j P_j$, $P_j = \xi_t^2 \cdots \xi_{t-j}^2$, $\|P_j\|_2 = 3^{(j+1)/2}$. If $\sqrt{3}\,\alpha_1 < 1$, Minkowski gives $\|y_t^2\|_2 \le \alpha_0 \sum_j \alpha_1^j 3^{(j+1)/2} < \infty$. Conversely, if $Ey_t^4 < \infty$, $m_4 := E\sigma_t^4$ is finite and constant, and with $Ey_{t-1}^4 = 3m_4$,

$$m_4 = E(\alpha_0 + \alpha_1 y_{t-1}^2)^2 = \alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1-\alpha_1} + 3\alpha_1^2 m_4, \quad \text{so} \quad (1-3\alpha_1^2)m_4 = \alpha_0^2 \frac{1+\alpha_1}{1-\alpha_1} > 0,$$

forcing $3\alpha_1^2 < 1$. Then $Ey_t^4 = 3m_4$ and

$$\frac{Ey_t^4}{(Ey_t^2)^2} = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)} \cdot \frac{(1-\alpha_1)^2}{\alpha_0^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3 \quad (\alpha_1 > 0).$$

Step 4. $\nu_t = \sigma_t^2(\xi_t^2 - 1)$ has $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E\nu_t^2 = E\sigma_t^4\, E(\xi_t^2-1)^2 < \infty$; a stationary square-integrable martingale difference is white noise ($E\nu_t\nu_{t+h} = E(\nu_t E(\nu_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$). Finally $y_t^2 = \sigma_t^2 + \nu_t = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t$. ■

Parts (1)–(2) need no normality. By (3) the marginal law is *leptokurtic*; e.g. 10,000 draws from $\sigma_t^2 = 0.2 + 0.8y_{t-1}^2$ give a histogram far more peaked and heavy-tailed than the normal.

Definition 6.2.1 (ARCH(q) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2, \tag{6.10}$$

with $\alpha_0 > 0$ and $\alpha_i \geq 0$ $(1 \leq i \leq q)$, then $\{y_t\}$ follows an *ARCH(q) model*. If moreover $1 - \alpha_1 z - \cdots - \alpha_q z^q \neq 0$ for all $|z| \leq 1$ (equivalently $\sum_{i=1}^q \alpha_i < 1$), then $\{y_t\}$ is strictly stationary and

$$\mathrm{Var}(y_t) = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i}. \tag{6.11}$$

Proof of the stationarity statement. [Supplement] Let $S = \sum_{i=1}^q \alpha_i$, $A(z) = \sum_i \alpha_i z^i$.

Step 1: the root condition is $S < 1$. If $S < 1$, $|A(z)| \leq S < 1$ on $|z| \leq 1$. If $S \geq 1$, $A(0) = 0$, $A(1) \geq 1$ give $A(z) = 1$ for some $z \in (0,1]$.

Step 2: construction. Let $h_t^{(0)} = 0$, $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^q \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)}$. By induction $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \dots) \geq 0$ is nondecreasing in m (nonnegative coefficients). As $h_{t-i}^{(m)} \perp \xi_{t-i}$, $Eh_t^{(m+1)} = \alpha_0 + S\, Eh_t^{(m)}$, so $Eh_t^{(m)} \leq \alpha_0/(1-S)$.

Step 3: limit. $\sigma_t^2 = \lim_m h_t^{(m)}$ has $E\sigma_t^2 = \alpha_0/(1-S)$ (monotone convergence) and satisfies $\sigma_t^2 = \alpha_0 + \sum_i \alpha_i \xi_{t-i}^2 \sigma_{t-i}^2$; with $y_t = \sigma_t \xi_t$ this is (6.10), strictly stationary as in Proposition , and $\mathrm{Var}(y_t) = E\sigma_t^2 E\xi_t^2$ gives (6.11).

Step 4: uniqueness. For another stationary solution $\tilde\sigma_t^2$ (function of past ξ, $E\tilde\sigma_t^2 < \infty$), $D_t = \sigma_t^2 - \tilde\sigma_t^2 = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i}$ gives $E|D_0| \leq S\, E|D_0|$, so $D_t = 0$ a.s. ■

As for ARCH(1), $\nu_t = y_t^2 - \sigma_t^2$ is a martingale difference and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2 + \nu_t,$$

so choosing q is choosing an AR order for $\{y_t^2\}$ (Chapter 4).

6.3 GARCH models

Motivation. Adding lagged conditional variances, as ARMA adds lagged noise to AR, gives a parsimonious alternative to high-order ARCH.

Definition 6.3.1 (GARCH(1,1) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_0 > 0,\ \alpha_1, \beta_1 \geq 0, \tag{6.12}$$

then $\{y_t\}$ follows a *GARCH(1,1) model*. If $\alpha_1 + \beta_1 < 1$, $\{y_t\}$ is (second-order) stationary; and if

$$E \log(\beta_1 + \alpha_1 \xi_t^2) < 0,$$

then $\{(y_t, \sigma_t)\}$ is strictly stationary (Nelson, 1990; Bougerol and Picard, 1992).

Proof of the stationarity statements. [Supplement] With $a(x) = \beta_1 + \alpha_1 x^2$, $\sigma_t^2 = \alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2$, so

$$\sigma_t^2 = \alpha_0\Big[1 + \sum_{k=1}^{K}\prod_{i=1}^{k} a(\xi_{t-i})\Big] + \prod_{i=1}^{K+1} a(\xi_{t-i})\,\sigma_{t-K-1}^2. \tag{6.13}$$

Step 1: strict stationarity. Let $\gamma = E\log a(\xi_t) < 0$ ($\le \log Ea < \infty$; if $\gamma = -\infty$ use $\max(\log a, -M)$). By the SLLN $\left(\prod_{i=1}^{k} a(\xi_{t-i})\right)^{1/k} \to e^{\gamma} < 1$, so by the root test

$$\sigma_t^2 := \alpha_0\Big[1 + \sum_{k=1}^{\infty}\prod_{i=1}^{k} a(\xi_{t-i})\Big]$$

converges a.s.; it satisfies $\alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2 = \sigma_t^2$, and (y_t, σ_t), a fixed function of the shifted i.i.d. sequence, is strictly stationary. Uniqueness: for another such $\tilde\sigma_t^2$, (6.13) gives $\tilde\sigma_t^2 - \sigma_t^2 = \prod_{i=1}^{K} a(\xi_{t-i})\,(\tilde\sigma_{t-K}^2 - \sigma_{t-K}^2)$; the product $\to 0$ a.s. and the second factor is tight, so the (K-free) left side is 0.

Step 2: second-order stationarity. If $\alpha_1 + \beta_1 < 1$, Jensen gives $E\log a(\xi_t) \le \log(\alpha_1 + \beta_1) < 0$, so Step 1 applies, and $E\prod_{i=1}^{k} a(\xi_{t-i}) = (\alpha_1 + \beta_1)^k$ gives

$$Ey_t^2 = E\sigma_t^2 = \alpha_0 \sum_{k\ge 0}(\alpha_1 + \beta_1)^k = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} < \infty;$$

with $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $\{y_t\}$ is stationary white noise. ∎

If $\alpha_1 + \beta_1 = 1$ and ξ_t^2 is nondegenerate, strict Jensen gives $E\log(\beta_1 + \alpha_1\xi_t^2) < 0$: strictly stationary with $Ey_t^2 = \infty$.

Fits to daily returns (e.g. $r_t = 100\log(P_{t+1}/P_t)$) typically give $\alpha_1 + \beta_1$ just below 1. $\alpha_1 + \beta_1 = 1$ is the *IGARCH model*; for $\alpha_1 + \beta_1 \approx 1$ the variance is *persistent*:

Proposition (Variance forecasts, [Supplement])

In the GARCH(1,1) model let $\phi = \alpha_1 + \beta_1$. For $k \ge 0$,

$$E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 \sum_{j=0}^{k-1}\phi^j + \phi^k\sigma_t^2, \tag{6.14}$$

which has the form $c + \phi^k\sigma_t^2$. If $\phi \approx 1$, the effect of today's variance on the forecast barely decays as k grows.

Proof. For $s \ge t$, as $\sigma_s^2 \in \mathcal{F}_{s-1}$ and $E(y_s^2 \mid \mathcal{F}_{s-1}) = \sigma_s^2$,

$$E(\sigma_{s+1}^2 \mid \mathcal{F}_{s-1}) = \alpha_0 + \alpha_1 E(y_s^2 \mid \mathcal{F}_{s-1}) + \beta_1\sigma_s^2 = \alpha_0 + \phi\sigma_s^2.$$

Induction on k with the tower property: $E(\sigma_{t+k+1}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 + \phi E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0\sum_{j=0}^{k}\phi^j + \phi^{k+1}\sigma_t^2$. ∎

6.3.1 Asymmetric models

Returns react more strongly to bad news.

Definition (Threshold GARCH(1,1))

Let $x^+ = \max\{0, x\}$ and $x^- = \max\{-x, 0\}$. If $y_t = \sigma_t \xi_t$ with $\{\xi_t\} \sim IID(0,1)$ and

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(y_{t-1}^+)^2 + \alpha_{12}(y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0,$$

then $\{y_t\}$ follows a *threshold* (or *asymmetric*) *GARCH(1,1) model*. Equivalently, $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \gamma_1 (y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2$ with $\alpha_1 = \alpha_{11}$, $\gamma_1 = \alpha_{12} - \alpha_{11}$.

$\alpha_{12} > \alpha_{11}$ (typical for stocks) means negative returns raise the next variance more.

6.3.2 Gaussian quasi-maximum likelihood

For $\xi_t \sim N(0,1)$, initial values y_0, σ_0^2 (commonly y_1^2) and $\theta = (\alpha_0, \alpha_1, \beta_1)^T$, compute

$$\sigma_t^{2*}(\theta) = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^{2*}(\theta), \qquad t = 1, \dots, n.$$

With $L(\theta) = \prod_{t=1}^n f(y_t \mid y_{t-1}, \dots)$,

$$f(y_t \mid y_{t-1}, \dots) = (2\pi\sigma_t^{2*})^{-1/2} \exp\{-y_t^2/(2\sigma_t^{2*})\},$$

the *Gaussian MLE* $\hat\theta$ maximizes

$$l(\theta) = \log L(\theta) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{t=1}^n \Big\{\log \sigma_t^{2*}(\theta) + \frac{y_t^2}{\sigma_t^{2*}(\theta)}\Big\}. \tag{6.15}$$

For non-normal ξ_t it is the *quasi-MLE* (QMLE), still consistent and asymptotically normal.

Proposition (Asymptotic normality of the QMLE, [Supplement])

Let $\{y_t\}$ be a strictly stationary GARCH(1,1) series with true parameter θ_0 in the interior of a compact parameter set, $\kappa_\xi = E\xi_t^4 < \infty$, and suppose $\hat\theta \to \theta_0$ in probability. Let $D_t = \sigma_t^{-2}\, \partial\sigma_t^2/\partial\theta$ at θ_0 and

$$J = E(D_t D_t^T) = -2\, E\Big[\frac{\partial^2 \log f(y_t \mid y_{t-1}, \dots)}{\partial\theta\, \partial\theta^T}\Big],$$

assumed positive definite. Then

$$\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, \Gamma), \qquad \Gamma = (\kappa_\xi - 1) J^{-1}$$

(Francq and Zakoian, 2004). For normal ξ_t, $\kappa_\xi = 3$ and $\Gamma = 2J^{-1}$.

Proof. [Supplement] Taken from Francq and Zakoian: initial values are negligible ($\sigma_t^{2*} - \sigma_t^2 = O(\beta_1^t)$), and D_t and second derivatives of $\log\sigma_t^2$ have uniformly bounded moments near θ_0. Let $l_t(\theta) = -\frac{1}{2}\{\log\sigma_t^2 + y_t^2/\sigma_t^2\}$.

Step 1: score. $\partial l_t/\partial\theta = \frac{1}{2}(y_t^2/\sigma_t^2 - 1)\sigma_t^{-2}\partial\sigma_t^2/\partial\theta$, at θ_0 equal to $s_t = \frac{1}{2}(\xi_t^2 - 1)D_t$, a stationary ergodic martingale difference ($D_t \in \mathcal{F}_{t-1}$) with $E(s_t s_t^T) = \frac{\kappa_\xi - 1}{4} J$; the martingale CLT gives $n^{-1/2}\sum_{t=1}^n s_t \xrightarrow{d} N(0, \frac{\kappa_\xi - 1}{4} J)$.

Step 2: Hessian.

$$\frac{\partial^2 l_t}{\partial\theta\, \partial\theta^T} = \frac{1}{2}\Big(\frac{y_t^2}{\sigma_t^2} - 1\Big)\frac{\partial}{\partial\theta^T}\Big(\sigma_t^{-2}\frac{\partial\sigma_t^2}{\partial\theta}\Big) - \frac{1}{2}\,\frac{y_t^2}{\sigma_t^2}\,\sigma_t^{-4}\frac{\partial\sigma_t^2}{\partial\theta}\frac{\partial\sigma_t^2}{\partial\theta^T}.$$

At θ_0, $E(\xi_t^2 \mid \mathcal{F}_{t-1}) = 1$, so its mean is $-\frac{1}{2}J$; by the ergodic theorem and the moment bound, $\frac{1}{n}\sum_t \partial^2 l_t(\bar\theta_n)/\partial\theta\partial\theta^T \xrightarrow{p} -\frac{1}{2}J$ for $\bar\theta_n \to \theta_0$.

Step 3: expansion. For an interior maximizer, $0 = n^{-1/2}\sum_t s_t + (\frac{1}{n}\sum_t \partial^2 l_t(\bar{\theta}_n))\sqrt{n}(\hat{\theta}-\theta_0)$, so $\sqrt{n}(\hat{\theta}-\theta_0) = 2J^{-1}n^{-1/2}\sum_t s_t + o_p(1)$, with covariance $2J^{-1}\cdot\frac{\kappa_\xi-1}{4}J\cdot 2J^{-1} = (\kappa_\xi-1)J^{-1}$. ■

Example 6.3.2 (Gaussian MLE for ARCH(1))

Given y_0, the conditional log-likelihood of ARCH(1) is

$$l(\alpha_0,\alpha_1) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{t=1}^{n}\Big\{\log(\alpha_0+\alpha_1 y_{t-1}^2) + \frac{y_t^2}{\alpha_0+\alpha_1 y_{t-1}^2}\Big\},$$

maximized numerically (no closed form).

6.3.3 GARCH(p,q)

Definition 6.3.3 (GARCH(p,q) model)

If $\{y_t\}$ satisfies

$$y_t = \xi_t\sigma_t, \quad \{\xi_t\}\sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q}\alpha_i y_{t-i}^2 + \sum_{j=1}^{p}\beta_j\sigma_{t-j}^2, \quad \alpha_0>0,\ \alpha_i,\beta_j\geq 0, \tag{6.16}$$

then $\{y_t\}$ follows a *GARCH(p,q) model.*

Proposition (Stationarity and ARMA form of GARCH(p,q), [Supplement])

Let $S = \sum_{i=1}^{q}\alpha_i + \sum_{j=1}^{p}\beta_j$.
(1) If $S<1$, (6.16) has a unique strictly stationary solution with σ_t^2 a function of $\xi_{t-1},\xi_{t-2},\ldots$ and $E\sigma_t^2<\infty$; it satisfies $Ey_t^2 = E\sigma_t^2 = \alpha_0/(1-S)$, and $\{y_t\}$ and $\{\sigma_t^2\}$ are stationary (Bollerslev, 1986). Conversely, a solution with $E\sigma_t^2$ finite and constant in t forces $S<1$.
(2) With $\nu_t = y_t^2-\sigma_t^2$, a martingale difference (white noise if $Ey_t^4<\infty$),

$$\Big(1-\sum_{i=1}^{q}\alpha_i B^i - \sum_{j=1}^{p}\beta_j B^j\Big)y_t^2 = \alpha_0 + \nu_t - \sum_{j=1}^{p}\beta_j\nu_{t-j}, \tag{6.17}$$

so $\{y_t^2\}$ follows an ARMA$(\max\{p,q\},p)$ model.

Proof.

Step 1: construction. $h_t^{(0)}=0$, $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^{q}\alpha_i\xi_{t-i}^2 h_{t-i}^{(m)} + \sum_{j=1}^{p}\beta_j h_{t-j}^{(m)}$. As for Definition 6.2.1, $h_t^{(m)}\geq 0$ is a nondecreasing function of past ξ's with $Eh^{(m+1)} = \alpha_0 + S\,Eh^{(m)}\uparrow\alpha_0/(1-S)$; the limit σ_t^2 solves (6.16), is strictly stationary, and $E(y_t\mid\mathcal{F}_{t-1})=0$ makes $\{y_t\}$ white noise.

Step 2: uniqueness. $D_t = \sigma_t^2-\tilde{\sigma}_t^2 = \sum_i\alpha_i\xi_{t-i}^2 D_{t-i} + \sum_j\beta_j D_{t-j}$ gives $E|D_0|\leq S\,E|D_0|$, so $D=0$.

Step 3: necessity. Expectations in (6.16) give $m=\alpha_0+Sm$ with $\alpha_0,m>0$, so $S<1$.

Step 4: ARMA form. $E(\nu_t\mid\mathcal{F}_{t-1})=0$ as in Proposition (4). Substituting $\sigma_{t-k}^2 = y_{t-k}^2-\nu_{t-k}$ $(k=0,\ldots,p)$ in (6.16),

$$y_t^2-\nu_t = \alpha_0 + \sum_{i=1}^{q}\alpha_i y_{t-i}^2 + \sum_{j=1}^{p}\beta_j(y_{t-j}^2-\nu_{t-j}),$$

which is (6.17), with AR degree $\max\{p,q\}$ and MA degree p. ■

So p, q are chosen by AIC/BIC (Chapter 4); GARCH(1,1) usually suffices.

Remark 6.3.4 (threshold AR(1)–GARCH(1,1))

A useful combination of conditional mean and asymmetric variance:

$$y_t = \phi y_{t-1} + \epsilon_t, \qquad \epsilon_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1),$$

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(\epsilon_{t-1}^+)^2 + \alpha_{12}(\epsilon_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0.$$

Heavy-tailed ξ_t (standardized t: *t-GARCH*) make the Gaussian QMLE less efficient, so testing normality matters (Section 6.4). A log-scale variant:

Definition 6.3.5 (EGARCH(1,1) model)

If $\{y_t\}$ satisfies $y_t = \sigma_t \xi_t$, $\{\xi_t\} \sim IID(0,1)$, and

$$\log \sigma_t^2 = \alpha_0 + \theta \frac{y_{t-1}}{\sigma_{t-1}} + \gamma\Big\{\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big| - E\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big|\Big\} + \beta \log \sigma_{t-1}^2,$$

where $\alpha_0, \theta, \gamma \in \mathbb{R}$ and $|\beta| < 1$, then $\{y_t\}$ follows an *EGARCH(1,1) model* (exponential GARCH).

No sign restrictions are needed; $\theta < 0$ means bad news raises volatility more. Stochastic volatility models are not treated.

6.4 Testing normality

Motivation. Non-normal GARCH errors cost the Gaussian MLE efficiency; Jarque–Bera detects them via skewness and kurtosis.

For i.i.d. $\xi_1, \ldots, \xi_n$ and H_0: $\xi_1, \ldots, \xi_n$ *are normal*, the *Jarque–Bera statistic* is

$$T_{JB} = n\Big(\frac{S_n^2}{6} + \frac{(K_n - 3)^2}{24}\Big), \qquad S_n = \frac{m_3}{m_2^{3/2}}, \quad K_n = \frac{m_4}{m_2^2}, \quad m_k = \frac{1}{n}\sum_{t=1}^{n}(\xi_t - \bar{\xi})^k, \tag{6.18}$$

with sample skewness $S_n \approx 0$ and kurtosis $K_n \approx 3$ under H_0.

Proposition (Null distribution of the Jarque–Bera statistic, [Supplement])

Under H_0, $T_{JB} \xrightarrow{d} \chi^2(2)$.

Proof. By invariance take $\xi_t \sim N(0,1)$; $\hat{\mu}_k = \frac{1}{n}\sum \xi_t^k$; $E\xi^2 = 1$, $E\xi^4 = 3$, $E\xi^6 = 15$, $E\xi^8 = 105$, odd moments 0.

Step 1: reduction. $\sqrt{n}\,\bar{\xi} = O_p(1)$, $\hat{\mu}_2 \to 1$, $\hat{\mu}_3 \to 0$ a.s., and

$$m_2 = \hat{\mu}_2 - \bar{\xi}^2, \quad m_3 = \hat{\mu}_3 - 3\bar{\xi}\hat{\mu}_2 + 2\bar{\xi}^3, \quad m_4 = \hat{\mu}_4 - 4\bar{\xi}\hat{\mu}_3 + 6\bar{\xi}^2\hat{\mu}_2 - 3\bar{\xi}^4,$$

so $\sqrt{n}\,m_3 = \sqrt{n}(\hat{\mu}_3 - 3\bar{\xi}) + o_p(1)$, $\sqrt{n}(m_4 - 3) = \sqrt{n}(\hat{\mu}_4 - 3) + o_p(1)$, $\sqrt{n}(m_2 - 1) = \sqrt{n}(\hat{\mu}_2 - 1) + o_p(1)$.

Step 2: delta method. $\sqrt{n}S_n = \sqrt{n}\,m_3 + o_p(1)$; $(x, y) \mapsto x/y^2$ has gradient $(1, -6)$ at $(3, 1)$, so $\sqrt{n}(K_n - 3) = \sqrt{n}(m_4 - 3) - 6\sqrt{n}(m_2 - 1) + o_p(1)$ and

$$\sqrt{n}\begin{pmatrix} S_n \\ K_n - 3 \end{pmatrix} = \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\begin{pmatrix} \xi_t^3 - 3\xi_t \\ \xi_t^4 - 6\xi_t^2 + 3 \end{pmatrix} + o_p(1).$$

Step 3: CLT. $\mathrm{Var}(\xi^3-3\xi)=15-18+9=6$, $\mathrm{Var}(\xi^4-6\xi^2)=105-180+108-9=24$, covariance an odd moment $=0$; so $\sqrt{n}S_n$, $\sqrt{n}(K_n-3)$ are asymptotically independent $N(0,6)$, $N(0,24)$ and $T_{JB}\xrightarrow{d}\chi^2(2)$. ■

For GARCH(1,1), residuals replace errors. Writing (6.12) as $(1-\beta_1 B)\sigma_t^2=\alpha_0+\alpha_1 y_{t-1}^2$,

$$\sigma_t^2=(1-\beta_1 B)^{-1}\alpha_0+\alpha_1(1-\beta_1 B)^{-1}y_{t-1}^2=\frac{\alpha_0}{1-\beta_1}+\alpha_1\sum_{j=1}^{\infty}\beta_1^{j-1}y_{t-j}^2,$$

and truncating with the Gaussian MLE,

$$\hat{\xi}_t=\frac{y_t}{\hat{\sigma}_t},\qquad \hat{\sigma}_t^2=\frac{\hat{\alpha}_0}{1-\hat{\beta}_1}+\hat{\alpha}_1\sum_{j=1}^{t-1}\hat{\beta}_1^{j-1}y_{t-j}^2,\qquad 1\le t\le n \tag{6.19}$$

(or the recursion (6.12) from $y_0=\sigma_0=y_1$; asymptotically equivalent). The residual statistic $\hat{T}_{JB}$ has the same $\chi^2(2)$ limit; reject if $\hat{T}_{JB}\ge c_\alpha$, $P(\chi^2(2)\ge c_\alpha)=\alpha$.

6.5 Change-point detection

Motivation. Ignoring a GARCH parameter shift typically yields a spurious IGARCH fit; the CUSUM of squared residuals detects it.

Test

H_0: $y_1,\dots,y_n$ follow a GARCH(1,1) model with no parameter change;
H_1: $y_1,\dots,y_n$ follow GARCH(1,1) but the parameters change at some time.

With residuals (6.19) from the Gaussian MLE under H_0, the *CUSUM statistic* is

$$\hat{T}_n=\max_{m\le k\le n}\frac{1}{\sqrt{n}\,\hat{\tau}}\Big|\sum_{t=1}^{k}\hat{\xi}_t^2-\frac{k}{n}\sum_{t=1}^{n}\hat{\xi}_t^2\Big|,\qquad \hat{\tau}^2=\frac{1}{n}\sum_{t=1}^{n}\hat{\xi}_t^4-\Big(\frac{1}{n}\sum_{t=1}^{n}\hat{\xi}_t^2\Big)^2, \tag{6.20}$$

$\hat{\tau}^2$ estimating $\tau^2=\mathrm{Var}(\xi_1^2)$, m small and fixed (5–10). Under H_0,

$$\hat{T}_n\approx T_n=\max_{m\le k\le n}\frac{1}{\sqrt{n}\,\hat{\tau}}\Big|\sum_{t=1}^{k}\xi_t^2-\frac{k}{n}\sum_{t=1}^{n}\xi_t^2\Big|, \tag{6.21}$$

small since both sums are $\approx k$. A change at k^* with $k^*/n\in(0,1)$ makes $\frac{1}{n}|\sum_{t\le k^*}\hat{\xi}_t^2-\frac{k^*}{n}\sum_{t\le n}\hat{\xi}_t^2|$ tend to a positive constant, so $\hat{T}_n\to\infty$.

Proposition (Limit of the CUSUM statistic, [Supplement])

Let ξ_t be i.i.d. with $E\xi_t^2=1$, $E\xi_t^4<\infty$, $\tau^2=\mathrm{Var}(\xi_t^2)>0$, and let $\hat{\tau}\to\tau$ in probability. Then

$$T_n\xrightarrow{d}T=\sup_{0\le u\le 1}|B^0(u)|,$$

where $B^0(u)=W(u)-uW(1)$ is a Brownian bridge and W a standard Brownian motion on $[0,1]$.

Proof.

Step 1. By Donsker, $W_n(u)=\frac{1}{\tau\sqrt{n}}\sum_{t\le\lfloor nu\rfloor}(\xi_t^2-1)\xrightarrow{d}W$ in $D[0,1]$.

Step 2. $\sum_{t\le k}\xi_t^2 - \frac{k}{n}\sum_{t\le n}\xi_t^2 = \sum_{t\le k}(\xi_t^2-1) - \frac{k}{n}\sum_{t\le n}(\xi_t^2-1)$, so with τ for $\hat\tau$ the statistic is $\max_{m\le k\le n}|W_n(k/n) - \frac{k}{n}W_n(1)|$, differing from $\sup_u|W_n(u)-uW_n(1)|$ by at most $|W_n(1)|/n + \max_{k<m}|W_n(k/n)| + m|W_n(1)|/n = o_p(1)$, as $\max_{k<m}|W_n(k/n)| \le \frac{1}{\tau\sqrt{n}}\sum_{t<m}|\xi_t^2-1| \to 0$.

Step 3. $x \mapsto \sup_u|x(u)-ux(1)|$ is continuous at continuous x, so $\sup_u|W_n(u)-uW_n(1)| \xrightarrow{d} T$; Slutsky with $\tau/\hat\tau \to 1$ gives $T_n \xrightarrow{d} T$. ■

T has the Kolmogorov law $P(T\le x) = 1 - 2\sum_{k\ge1}(-1)^{k-1}e^{-2k^2x^2}$, $P(T\ge 1.358)=0.05$: reject at 5% if $\hat T_n \ge 1.358$, and estimate the change point by

$$\hat k = \arg\max_{1\le k\le n}\hat T_n(k), \qquad \hat T_n(k) = \frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{j=1}^{k}\hat\xi_j^2 - \frac{k}{n}\sum_{j=1}^{n}\hat\xi_j^2\Big|. \tag{6.22}$$

Binary segmentation (retesting $y_1,\dots,y_{\hat k}$ and $y_{\hat k+1},\dots,y_n$) finds several changes. Power is highest for changes near $n/2$. An apparent IGARCH fit ($\hat\alpha_1+\hat\beta_1\approx 1$) should prompt a change-point test; many stock series show changes.

6.6 Value-at-Risk

Motivation. VaR is a low conditional quantile, in GARCH the volatility times an error quantile, estimable by quantile regression.

For X with d.f. F_X and $0<\tau<1$, the *τ-quantile* is

$$F_X^{-1}(\tau) = \min\{y : F_X(y)\ge\tau\}$$

(attained by right-continuity).

Proposition (Two properties of quantiles, [Supplement])

(1) For $a>0$ and $b\in\mathbb{R}$,

$$F_{aX+b}^{-1}(\tau) = aF_X^{-1}(\tau)+b. \tag{6.23}$$

(2) If $E|X|<\infty$ and $\rho_\tau(x) = x(\tau - I(x<0))$ (the *check function*), then

$$F_X^{-1}(\tau) = \min\Big\{m:\ m \text{ minimizes } E\rho_\tau(X-m)\Big\}. \tag{6.24}$$

Proof.

Step 1. $F_{aX+b}(ax+b) = F_X(x)$ and $x\mapsto ax+b$ is an increasing bijection, so

$$\min\{y : F_{aX+b}(y)\ge\tau\} = \min\{ax+b : F_X(x)\ge\tau\} = a\min\{x : F_X(x)\ge\tau\}+b.$$

Step 2. $\rho_\tau(x-m') - \rho_\tau(x-m) = \int_m^{m'}(I(x<u)-\tau)du$ for $m<m'$; by Fubini, with $g(m) = E\rho_\tau(X-m)$ (finite as $\rho_\tau(x)\le|x|$),

$$g(m')-g(m) = \int_m^{m'}(P(X<u)-\tau)du = \int_m^{m'}(F_X(u)-\tau)du$$

(integrands differ only at countably many atoms). With $q = F_X^{-1}(\tau)$: for $m<q$, $F_X<\tau$ on $[m,q)$, so $g(q)<g(m)$; for $m>q$, $F_X\ge\tau$ on $[q,m]$, so $g(m)\ge g(q)$. ■

So the sample quantile minimizes

$$\frac{1}{n}\sum_{t=1}^{n}\rho_\tau(X_t-m). \tag{6.25}$$

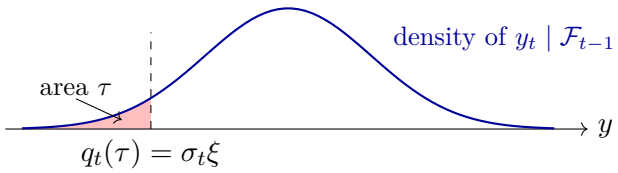


Figure 2: Conditional VaR: the τ-quantile of $y_t \mid \mathcal{F}_{t-1}$, scaling with σ_t.

Reparametrize GARCH(1,1) with the scale in the error variance ω:

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim iid(0,\omega),\ \omega > 0, \qquad \sigma_t^2 = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \alpha, \beta \geq 0,\ \alpha + \beta < 1.$$

As $\sigma_t \in \mathcal{F}_{t-1}$ and $\xi_t \perp \mathcal{F}_{t-1}$, (6.23) gives the *conditional quantile*

$$q_t(\tau) = \sigma_t F_\xi^{-1}(\tau) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2)^{1/2}\, \xi, \qquad \xi = F_\xi^{-1}(\tau),$$

a function $q_t(\xi, \alpha, \beta)$, and by the conditional (6.24)

$$q_t(\tau) = \min\{m :\ m \text{ minimizes } E(\rho_\tau(y_t - m) \mid \mathcal{F}_{t-1})\}.$$

As in (6.25), $(\hat{\xi}, \hat{\alpha}, \hat{\beta})$ minimizes

$$\frac{1}{n} \sum_{t=1}^{n} \rho_\tau(y_t - q_t^*(\xi, \alpha, \beta)), \qquad q_t^*(\xi, \alpha, \beta) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*})^{1/2} \xi,$$

with $\sigma_t^{2*} = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*}$ from arbitrary y_0, σ_0^{2*}. Then

$$\hat{q}_t(\tau) = q_t^*(\hat{\xi}, \hat{\alpha}, \hat{\beta}) =: \mathrm{VaR}_{Y_t \mid \mathcal{F}_{t-1}}(\tau)$$

is the *conditional Value-at-Risk* (Figure 2), with small τ such as 0.01 or 0.05.

6.7 Selected exercises

Motivation. Exercises proving the conditional law, uniqueness and strict stationarity used in the ARCH(1) construction.

Exercise 6.1 (Conditional law of ARCH(1))

Let X, Y be independent random elements (with values in $\mathbb{R}^k$ or $\mathbb{R}^\infty$) and f a measurable real function. It is known that

$$P(f(X,Y) \leq z \mid X = x) = P(f(x,Y) \leq z).$$

Prove this identity, and use it to prove (6.3).

Solution.

Step 1: the identity. Let $g(x) = P(f(x,Y) \leq z) = \int \mathbb{1}\{f(x,u) \leq z\}\, P_Y(du)$. As the law of (X,Y) is $P_X \otimes P_Y$, Tonelli gives g measurable and

$$P(f(X,Y) \leq z,\ X \in A) = \int_A \int \mathbb{1}\{f(x,u) \leq z\}\, P_Y(du)\, P_X(dx) = E[g(X)\mathbb{1}\{X \in A\}],$$

the defining property of $P(f(X,Y) \leq z \mid X = x)$.

Step 2: application. $y_{t-1} \perp \xi_t$ (Proposition (1)); with $X = y_{t-1}$, $Y = \xi_t$, $f(x,u) = u\,(\alpha_0+\alpha_1 x^2)^{1/2}$,

$$P(y_t \le z \mid y_{t-1} = y) = P(\xi_t(\alpha_0 + \alpha_1 y^2)^{1/2} \le z) = \Phi(z/\sqrt{\alpha_0 + \alpha_1 y^2}),$$

which is (6.3).

Step 3: the whole past. With $X = (y_{t-1}, y_{t-2}, \dots) \perp \xi_t$ and $f(x,u) = u(\alpha_0 + \alpha_1 x_1^2)^{1/2}$, $y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$. ■

Exercise 6.2 (Iterating the ARCH(1) recursion)

Consider the ARCH(1) model (6.2).
(a) Prove (6.4) for every $n \ge 0$.
(b) Let $\{y_t\}$ be a solution with $Ey_t^2 = c < \infty$ for all t, where each y_s is a function of $\xi_s, \xi_{s-1}, \dots$. Prove that y_t^2 and σ_t^2 are given by the series (6.5); consequently y_t equals (6.6).

Solution.

(a) Let $P_{t,j} = \xi_t^2 \cdots \xi_{t-j}^2$. For $n = 0$, $y_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2 = \alpha_0 P_{t,0} + \alpha_1 P_{t,0}\, y_{t-1}^2$. Induction: substituting $y_{t-n-1}^2 = (\alpha_0 + \alpha_1 y_{t-n-2}^2)\xi_{t-n-1}^2$,

$$\alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = \alpha_0 \alpha_1^{n+1} P_{t,n+1} + \alpha_1^{n+2} P_{t,n+1}\, y_{t-n-2}^2.$$

(b)

Step 1. With $S_n = \alpha_0 \sum_{j=0}^{n} \alpha_1^j P_{t,j}$ and $R_n = \alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = y_t^2 - S_n$, $R_n \ge 0$ is nonincreasing, so $R_n \downarrow R \ge 0$ a.s.

Step 2. $y_{t-n-1} \perp (\xi_t, \dots, \xi_{t-n})$, so

$$ER_n = \alpha_1^{n+1} E(\xi_t^2) \cdots E(\xi_{t-n}^2)\, Ey_{t-n-1}^2 = \alpha_1^{n+1} c \to 0;$$

hence $R = 0$ a.s. and $y_t^2 = \alpha_0 \sum_{j \ge 0} \alpha_1^j P_{t,j}$.

Step 3. At time $t-1$, with $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \alpha_0 \sum_{j \ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-1-j}^2,$$

and $y_t = \sigma_t \xi_t$ is (6.6): with Proposition (1), the *only* causal finite-variance solution. ■

Exercise 6.4 (Functions of an i.i.d. sequence)

Let $\{\xi_t\}$ be an i.i.d. sequence.
(a) For measurable $f_n : \mathbb{R}^n \to \mathbb{R}$, show that $Y_t^{(n)} = f_n(\xi_t, \dots, \xi_{t-n+1})$ is strictly stationary (in particular for $n = 2$).
(b) Let $f : \mathbb{R}^\infty \to \mathbb{R}$ be measurable and suppose $f_n(\xi_t, \dots, \xi_{t-n+1}) \to f(\xi_t, \xi_{t-1}, \dots)$ a.s. for every t. Show that $Y_t = f(\xi_t, \xi_{t-1}, \dots)$ is strictly stationary.

Solution.

(a) For times $t_1, \dots, t_m$, shift h, $L = \min_i t_i - n + 1$, $U = \max_i t_i$,

$$(Y_{t_1}^{(n)}, \dots, Y_{t_m}^{(n)}) = g(\boldsymbol{\xi}), \qquad (Y_{t_1+h}^{(n)}, \dots, Y_{t_m+h}^{(n)}) = g(\boldsymbol{\xi}^h),$$

with the same measurable g, $\boldsymbol{\xi} = (\xi_L, \dots, \xi_U)$, $\boldsymbol{\xi}^h = (\xi_{L+h}, \dots, \xi_{U+h})$. Both have law $\mu^{\otimes(U-L+1)}$, so the images agree in law (Definition 1.2.2); $n = 2$ is a special case.

(b) $\mathbf{V}_n = (Y_{t_1}^{(n)}, \dots, Y_{t_m}^{(n)}) \to \mathbf{V}$ and $\mathbf{V}_n^h \to \mathbf{V}^h$ a.s., hence in distribution; by (a) $\mathbf{V}_n \overset{d}{=} \mathbf{V}_n^h$, so by uniqueness of weak limits $\mathbf{V} \overset{d}{=} \mathbf{V}^h$. Taking f_n the truncation of (6.6) proves the strict stationarity in Proposition (1). ■